Comprehensive Prioritization Framework for AI-Driven Discovery of Novel Antimicrobial Agents from Medicinal Plants

Executive Summary

Antimicrobial resistance (AMR) poses a critical global health threat, and medicinal plants – rich in diverse secondary metabolites – are a historically proven source of new antimicrobials[1][2]. This framework leverages phytochemical diversity, traditional knowledge, and modern bioassays to prioritize plant leads. Plants with high concentrations of known antimicrobial classes (e.g., flavonoids, alkaloids, terpenoids, quinones) are emphasized, since these compounds "are a major source of chemical diversity" and "promising sources of untapped potent antibacterial agents"[2]. Likewise, ethnobotanical evidence is scored: broad cultural consensus and long history of use correlate with likely efficacy[3][4]. The approach then integrates quantitative potency data (e.g. MIC values, bactericidal vs. static activity), coverage of resistant pathogens (especially WHO's ESKAPE priority bacteria[5]), synergistic potential with existing drugs, and endophyte genomic insights. A multi-criteria decision analysis (MCDA) algorithm combines all metrics into an overall priority score[6]. In summary, this standardized workflow – from data curation through AI-driven ranking – focuses resources on the most promising plants, ensuring scientific rigor and compliance while maximizing the chance of finding novel, potent antimicrobials in nature[1][3].

1.0 Phytochemical Profiling and Secondary Metabolite Assessment

Rationale: Plants produce specialized secondary metabolites as defenses against microbes. Profiling these compounds guides prioritization because "plants are rich in secondary metabolites and are a major source of chemical diversity"[2]. Broad metabolite classes (flavonoids, terpenes, alkaloids, etc.) often have antibacterial activity.

1.1 Secondary Metabolite Classification and Scoring

- **Compound Class Prioritization**: Give top priority to plants containing high levels of classes known for antimicrobial effects (e.g. alkaloids, terpenoids, polyphenols, quinones, antimicrobial peptides)[2]. For example, phenolic compounds can damage bacterial membranes or interfere with metabolism, while alkaloids and terpenoids often disrupt enzymes or cell structures. In practice, "active groups" identified by phytochemical analysis – such as flavonoids, tannins, alkaloids, terpenes and coumarins – are marked as high-value[2]. Plants meeting concentration thresholds (e.g. >0.1% dry weight for rich metabolites) and having multiple structural subclasses in a class are favored.
- **Quantitative Scoring**: A weighted score can combine metabolite class abundance, structural diversity, and concentration. For example, assign higher weights to rare but potent classes (alkaloids, terpenoids) and multiply by a diversity index (number of distinct scaffolds) and a normalized concentration factor. This yields a

Phytochemical_Score that ranks extracts by expected potency. (Such formulas build on metabolomic scoring concepts, though specific weights are context-dependent.)

- **Structural Complexity:** Within candidate compounds, greater molecular complexity often correlates with novelty and potential to evade resistance. Metrics like the Bertz complexity index or counting stereocenters can be used. Highly complex molecules (many rings, chiral centers, >3 stereocenters) that are dissimilar (Tanimoto similarity <0.7) to known antibiotics are especially prized. These features suggest "novelty to human pathogens" and the potential to "combat multidrug-resistant pathogens"[7]. Reactive functional groups (e.g. electrophiles, alkylators) and natural product-like scaffolds (but not generic primary metabolites) further enhance interest.

1.2 Metabolic Pathway Analysis

- **Biosynthetic Pathways:** Analyze enriched metabolic pathways using databases (KEGG, PlantCyc) and predictive tools (e.g. antiSMASH on plant/endophyte genomes). Plants whose genomes encode pathways for antimicrobial natural products (e.g. polyketides, nonribosomal peptides) may yield novel compounds. Consider whether stress conditions (drought, infection) upregulate these pathways – seasonal or ecological data can hint which environmental triggers boost active metabolites.
- **Gene Cluster Mining**: For plants with available genomic or transcriptomic data, identify biosynthetic gene clusters (BGCs) for secondary metabolites. Clusters unique to a plant or with rare domains suggest uncharacterized chemistry. Horizontal transfer of BGCs (e.g. from endophytes) may indicate unique metabolites. In summary, pathways enabling antibiotic-like compounds are prioritized, especially if correlated with high metabolite production under natural conditions.

2.0 Ethnobotanical Significance and Traditional Knowledge Validation

Rationale: Empirical traditional use provides in vivo validation of safety and efficacy. A plant used widely across cultures for infections is statistically more likely to contain a real antimicrobial. Quantitative ethnobotanical metrics (use value, consensus factor) help formalize this evidence[4][3].

2.1 Multi-Dimensional Ethnobotanical Scoring

- **Ethnobotanical Score:** Integrate factors reflecting traditional endorsement. One approach is a weighted formula:
- **Cultural Consensus:** Number of independent cultural systems using the plant medicinally (high consensus → higher score). For instance, a high Informant Consensus Factor (ICF) indicates homogenous ethnomedicinal knowledge among respondents[4].
- **Geographic Distribution:** Use across multiple biogeographic regions or countries. A broad distribution suggests convergent validation of the remedy.
- **Historical Duration:** Length of documented use in folk medicine. Centuries-old use implies time-tested reliability.
- **Preparation Specificity:** Detail in traditional preparations (precise dosage, method, combination). More specific recipes imply careful empirical optimization.

Mathematically, one can combine these (e.g. weighted sum) to yield an Ethnobotanical_Score. Note that ethnobotanical literature commonly uses similar metrics: for example, ICF itself is a score of use agreement[4], and high ICF values correlate with effective medicinal plants.

- **Traditional Preparation Analysis:** Examine ethnobotanical reports for extraction and formulation methods. Different solvents (water decoction, ethanol tincture, oil infusion, vinegar maceration) can extract different phytochemicals, guiding laboratory extraction choices. Complex processes (fermentation, charring, multi-plant combinations) may indicate attempts to enhance potency or reduce toxicity. Routes of administration (topical poultice vs. oral tea vs. inhalation) also provide clues about expected bioactivity. These specifics can be noted but need no formal score; however, a plant with a well-documented, replicable preparation is more convincing.

2.2 Cultural and Geographical Validation

- **Cross-Verification:** Confirm reported uses by at least three independent sources (different regions or ethnic groups). Independent verification strengthens confidence. For example, if the same plant is used for treating wounds in East Asia, Africa, and South America, it suggests a true pharmacological effect. Such cross-cultural “hits” can be noted as evidence of broad efficacy.
- **Environmental Correlation:** In some cases, link plant use to ecology. For instance, plants used in arid zones (where infections were historically deadly) might have evolved potent defenses. Also, certain communities use particular herbs only during seasonal disease outbreaks, indicating knowledge of compound fluctuations. While hard to quantify, these contextual details enrich the ethnobotanical profile.
- **Conservation of Knowledge:** Finally, engage traditional healers (with consent) for insights. This ensures ethical sourcing of data and may reveal unpublished details (specific dosages, plant parts, synergistic combinations) that inform laboratory work. Respect for knowledge holders and benefit-sharing agreements are integral to this process, aligning with WHO guidelines on traditional medicine[3].

3.0 Inhibitory Activity and Potency Metrics

Rationale: Empirical potency screening is essential. Initial bioassays confirm that prioritization has practical validity (i.e., the extract truly inhibits pathogens) and guide chemical isolation steps. Standardized potency criteria (MIC, MBC) categorize candidates by strength.

3.1 Comprehensive Antimicrobial Activity Profiling

- **Tiered Testing Protocol:** Adopt a stepwise bioassay scheme:
- **Crude Extract Screening (Tier 1):** Test whole plant extracts (e.g. methanol, ethanol, aqueous) against target bacteria to determine MIC (minimum inhibitory concentration). A common cutoff is MIC <100 µg/mL for crude extracts to be considered significantly active, although this can vary by study. Hits are advanced for fractionation.

- **Bioactivity-Guided Fractionation (Tier 2):** Divide active extracts into fractions (by solvent partitioning or chromatography) and re-test each fraction. Fractions that retain activity indicate the presence of the bioactive compound(s).
- **Pure Compound Evaluation (Tier 3):** Isolate individual compounds from active fractions and determine their MICs. Pure compounds with MIC <10 µg/mL (or <10 µM) against pathogens are highly promising.
- **Mechanism Studies (Tier 4):** For pure actives, conduct target-based assays (e.g. enzyme inhibition, cell imaging) to elucidate the mode of action. Finding a novel target mechanism is especially valuable for AMR.
  This tiered approach is widely used in natural product discovery, analogous to protocols in other screening programs[7].
- **Bactericidal vs. Bacteriostatic:** Determine if an extract/compound kills bacteria (bactericidal) or merely inhibits growth (bacteriostatic) by measuring the MBC (minimum bactericidal concentration) in relation to the MIC. Typically, an MBC/MIC ratio ≤4 indicates bactericidal activity (desirable for serious infections), whereas a higher ratio implies bacteriostatic action. This classification informs priority; bactericidal agents are often weighted higher. (Technical note: exact cutoffs can vary, but the ≤4 rule is a common convention.)
- **Advanced Potency Metrics:** For candidates passing Tier 3, conduct additional assays: time-kill kinetics (plotting bacterial viability over 24 hours to see how rapidly cells are killed), post-antibiotic effect duration (time cells remain suppressed after drug removal), and dose-response curves to calculate IC_{50}. Also assess cytotoxicity (CC_{50} on mammalian cells) to compute a therapeutic index (CC_{50}/MIC). A high therapeutic index (>10 or more) indicates selectivity for microbes over host cells. These metrics guide go/no-go decisions for further development.

3.2 Reference Standard Comparison

Always benchmark activity against known controls. Use reference antibiotics (e.g. ampicillin, ciprofloxacin) to contextualize MIC values. Likewise, test known antimicrobial phytochemicals (e.g. berberine, allicin) as natural comparators. For example, berberine from Coptis spp. not only has intrinsic activity but also synergizes with antibiotics: it can lower the MICs of azithromycin or levofloxacin by 50–97%[8]. Documenting where a plant extract stands relative to these standards helps prioritize truly exceptional candidates. A selectivity index (ratio of host cell IC_{50} to bacterial MIC) should also be calculated here to flag any cytotoxicity concerns.

4.0 Activity Against Resistant Pathogens

Rationale: Focusing on multi-drug resistant (MDR) pathogens ensures relevance to AMR. The WHO's top targets are the ESKAPE pathogens (Enterococcus faecium, S. aureus including MRSA, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacter species)[5]. A promising plant agent should ideally be active against at least several of these.

4.1 ESKAPE Pathogen Panel Testing

- **Critical Priority:** Test extracts/compounds against MDR "critical" strains: carbapenem-resistant A. baumannii, vancomycin-resistant E. faecium (VRE), and MRSA. These often carry multiple resistance mechanisms (enzymes, efflux pumps, target mutations)[9], so activity here is a high bar.
- **High/Medium Priority:** Also include extended-spectrum β-lactamase (ESBL) K. pneumoniae, multidrug-resistant P. aeruginosa, and resistant E. coli or S. pneumoniae. Assign weight factors reflecting clinical threat (e.g., 1.0 for critical, 0.8 for high, 0.6 for medium).
- **Coverage Score:** Calculate a resistance coverage score (e.g. number of resistance mechanisms overcome divided by total known for that pathogen). Emphasize agents active against multiple mechanisms (e.g. both ESBLs and efflux-pump-driven strains). Incorporate anti-biofilm activity here, since many resistant infections involve biofilms.

4.2 Novel Resistance Mechanism Targeting

- **Membrane Disruptors:** Identify compounds that target bacterial membranes or cell wall precursors (e.g. lipid II). Such mechanisms (like those of daptomycin or certain antimicrobial peptides) remain effective even when common drug targets are mutated. Plant-derived peptides or lipophilic terpenoids often act this way.
- **Efflux Pump Inhibitors:** Some phytochemicals can block efflux pumps, restoring antibiotic activity. For instance, piperine (from black pepper) has been shown to inhibit S. aureus efflux pumps and thereby potentiate ciprofloxacin[10]. Screening for efflux inhibition (e.g. by fluorometric assays) identifies such adjuvant candidates.
- **Biofilm and Persister Targeting:** Biofilms harbor dormant "persister" cells shielded from drugs. Compounds that penetrate biofilms or kill persisters are valuable. For example, clove oil's eugenol disrupts S. aureus biofilms and suppresses toxin production[11]. Any activity that shortens biofilm survival or re-sensitizes persisters should be flagged. Transcriptomic profiling of treated biofilms can reveal suppression of virulence genes or stress responses, guiding further focus.
- **Virulence Suppression:** Beyond killing bacteria, some phytochemicals attenuate virulence factor expression. Eugenol (mentioned above) inhibits bacterial toxin genes[11], and other molecules (e.g. some flavonoids) can block quorum sensing. Lower virulence reduces pathogenicity even if bacteria persist. Such mechanisms broaden the impact of discovery beyond classical bactericidal action.

5.0 Synergism and Resistance Modifying Activity (RMA)

Rationale: Combination therapies are key to combating AMR. Plant compounds can work synergistically with each other or with antibiotics, restoring efficacy. Synergistic hits are prioritized as they offer immediate translational potential.

5.1 Synergy Screening Framework

- **Checkerboard Assays:** Test combinations of plant extract (or isolated compounds) with standard antibiotics using checkerboard dilution to calculate the fractional inhibitory concentration index (ΣFIC). Interpret ΣFIC as: ≤0.5 = strong synergy; 0.5–1.0 = additive; 1.0–4.0 = indifference; >4 = antagonism. Prioritize pairs with ΣFIC ≤0.5 for further study. For example, berberine combined with azithromycin or levofloxacin has dramatically lowered MICs (ΣFIC indicative of strong synergy)[12].
- **Efflux Pump Inhibition:** Evaluate if the plant extract itself can reduce antibiotic MIC by efflux inhibition. Quantify fold-decrease in MIC of a test antibiotic in presence of subinhibitory plant concentrations. A ≥4-fold reduction is considered strong efflux inhibition, consistent with synergy. (E.g. piperine yielded >4× potentiation of ciprofloxacin against S. aureus[10].)
- **Interpretation and Prioritization:** Assign a Synergy_Score based on the best ΣFIC achieved with any antibiotic. ΣFIC ≤0.5 combinations get highest sub-scores. Agents with broad synergy (multiple antibiotics or herb-pairs) rank higher. Antagonistic interactions should exclude candidates (ΣFIC >4 suggests the plant interferes with antibiotics).

5.2 Multi-Mechanism Synergy Profiling

- **Time-Kill Synergy:** For strong hits (ΣFIC ≤0.5), perform time-kill assays with drug combinations to confirm synergy in killing kinetics over 24–48 hours. A >2 log₁₀ reduction in CFU (over best single drug) is typically considered synergy in kill curves.
- **Resistance Prevention:** Test for development of resistance under serial passage with the extract or combination. Synergistic combinations often suppress resistance emergence. For example, combining an efflux inhibitor with an antibiotic can delay or prevent target-based resistance.
- **Omics Analysis:** Use transcriptomics or proteomics on bacteria treated with synergistic combos to elucidate combined effects (e.g. downregulated stress responses). Such data, while exploratory, can validate that the combination hits multiple pathways and thus may prevent resistance.

6.0 Endophyte Correlation and Holistic Activity Assessment

Rationale: Endophytic microbes (fungi/bacteria living inside plants) often produce the same or similar bioactive metabolites as their hosts[13]. Considering the endophytic microbiome can reveal new sources of the same compounds or novel compounds not made by the plant alone.

6.1 Endophytic Microbiome Analysis

- **Isolation & Sequencing:** Surface-sterilize plant tissues and culture endophytic bacteria/fungi. Perform DNA extraction and 16S/ITS sequencing of plant tissues to

profile endophyte diversity. Isolates should be taxonomically identified (e.g. via rRNA sequencing).

- **Bioactivity Screening:** Test isolated endophytes (culture extracts) for antimicrobial activity. Sometimes an endophyte alone produces the active compound. Compare activity profiles: if a plant extract and a certain endophyte share activity against the same pathogens, they may share chemistry.
- **Correlation Scoring:** Calculate a correlation (e.g. Pearson R) between plant extract potency and presence/diversity of endophytes. A high correlation (R ≥0.7) suggests the microbiome contributes to bioactivity and should increase the plant's priority. This correlation is an independent score in the overall algorithm.

6.2 Biosynthetic Gene Cluster (BGC) Mining

- Metagenomic Analysis: Sequence the metagenome of plant tissues or endophyte cultures. Use tools like antiSMASH to scan for BGCs encoding known antimicrobial types (NRPS, PKS, terpene synthases, etc.). The presence of novel BGCs (with low similarity to known clusters) indicates potential new chemistry.
- Horizontal Gene Transfer: Look for plant genes that resemble microbial BGC enzymes (e.g. PKS or cytochrome P450s not typical of the plant lineage). This may suggest past gene transfer from endophytes.
- Comparative Genomics: If multiple populations of the same plant species are available, compare their endophyte BGC content. Unique clusters in high-priority populations (e.g. from stressed habitats) can pinpoint special chemotypes. Overall, evidence that endophytes and plant share biosynthetic capabilities boosts confidence in that plant as a source.

7.0 Integrated Prioritization Algorithm

After gathering all data, a multi-criteria engine computes an overall score for each plant. This Decision Engine systematically combines the sub-scores with defined weights (reflecting expert judgment or learned importance).

7.1 Multi-Criteria Decision Engine

The overall Priority Score is calculated as a weighted sum of normalized criteria: phytochemical richness, ethnobotanical score, potency, resistant pathogen coverage, synergy, and endophyte correlation. Using methods from multi-criteria decision analysis (MCDA), each criterion is scaled 0–100 and weighted (e.g. 25% phytochemistry, 20% ethnobotany, etc.)[6]. MCDA is commonly used in drug discovery to "systematically prioritize… various drug properties based on their relative importance"[6]. By integrating diverse data (chemical, biological, cultural) in this way, the algorithm objectively ranks candidates.

Priority Bands: Map the final score to action bands: e.g. P1 (critical, score 85–100) means immediate testing with maximal resources, P2 (70–84) means high priority for follow-up, P3

(50–69) is medium priority, and P4 (<50) is low priority for archival or future reconsideration. This helps allocate limited lab and field resources efficiently.

Priority Band	Score Range	Action	Resource Allocation
P1 – Critical	85–100	Immediate testing	Maximum (dedicated team)
P2 – High	70–84	Short-term testing	Significant
P3 – Medium	50–69	Batch testing	Moderate
P4 – Low	<50	Archive / re-evaluate	Minimal

7.2 AI-Enhanced Prediction Architecture

Modern AI and machine learning can automate score prediction. For example, a class like AMRCandidatePrioritizer could load pre-trained models for each criterion: a model that predicts phytochemical richness from genomic/metabolomic data, an ethnobotany model (from text mining sources), ML-based potency predictors, and so on. Each sub-score is predicted and multiplied by its weight to yield the final priority. In practice, deep learning (e.g. graph neural nets) is being used to predict antimicrobial activity of compounds[14], and MCDA frameworks like AHP or TOPSIS can operationalize the weighted scoring[6]. The pseudocode below illustrates this integration:

```
class AMRCandidatePrioritizer:
    def __init__(self, trained_models, criteria_weights):
        self.models = trained_models
        self.weights = criteria_weights

    def calculate_priority_score(self, plant_data):
        scores = {
            'phytochemical': self.calculate_phytochemical_score(plant_data),
            'ethnobotanical': self.calculate_ethnobotanical_score(plant_data),
            'potency': self.predict_potency_score(plant_data),
            'resistance': self.predict_resistance_coverage(plant_data),
            'synergy': self.predict_synergy_potential(plant_data),
            'endophyte': self.predict_endophyte_correlation(plant_data)
        }
        priority_score = sum(scores[c] * self.weights[c] for c in scores)
        return priority_score, scores
```

Fig: Schematic Python class combining AI models to compute weighted priority scores.

8.0 Implementation and Quality Control

- **Validation:** For top-ranked (P1) plants, conduct inter-laboratory validation. Share samples and data blind between independent labs to ensure reproducibility of extraction and bioassay results. Standardize extraction protocols (solvent, time, plant part) and include positive/negative controls in every assay. Also test each promising candidate in multiple seasons or harvest times to verify consistent bioactivity (seasonal reproducibility).
- **Controls and Blinding:** Always include known-active extracts and media-only blanks in every assay. Where feasible, perform double-blind testing (lab technicians not knowing sample identity) to remove bias.
- **Documentation:** Record all methods and raw data meticulously. Deposit sequence data (for genomes or microbiomes) in public databases and log metadata (location, collection date, voucher specimen ID) to support future follow-up.
- **Ethics and Compliance**: Ensure collection permits and benefit-sharing agreements are in place for all plant samples and traditional knowledge. Consult local stakeholders and ethical review boards, especially when working with indigenous knowledge.

9.0 Operational Workflow

Phase	Duration	Key Activities
1. Data Collection & Curation	Weeks 1–2	Assemble a comprehensive plant database (phytochemicals, usage, genomes). Ensure data quality and prepare datasets for AI models.
2. AI Prediction & Scoring	Weeks 2–4	Run prioritization algorithm on all candidates. Generate scores and rank plants. Perform sensitivity analysis on weights.
3. Wet Lab Validation Planning	Weeks 4–6	Design bioassay strategy for top candidates (e.g. MIC testing panels, fractionation plan). Allocate resources (reagents, personnel).
4. Continuous Learning	Ongoing	Incorporate new assay results into the dataset, retrain models monthly, adjust criteria weights quarterly, and update plans annually.

This framework is intended as a dynamic, iterative guide: as new data (e.g. novel assays, resistance trends, or ethnobotanical discoveries) emerge, the pipeline adapts. By uniting traditional knowledge with modern analytics and rigorous testing, it maximizes the chance of uncovering breakthrough antimicrobials from nature[1][3].

Sources: The above recommendations synthesize findings from recent reviews and studies on natural antimicrobials and drug discovery (see, e.g., Hayden et al. 2022[2][7][8][15]; de Lima Procópio et al. 2025[1][16]; Yilmaz et al. 2023[3]; Wang et al. 2024[17]; Bachorz et al. 2025[6]). All criteria and thresholds are based on community standards and literature consensus in antimicrobial natural products research.

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